Object Oriented Programming (CSEG1044)

Unit 04 – Lecture 04: Swing Basics (Components, Containers, Layouts)

April 1, 2026

Sources: Schildt, *Java: The Complete Reference* (12e) (generics, lambdas, Swing, JDBC).; Bloch, *Effective Java* (3e) (generics + lambdas).; Oracle Java Tutorials: Generics + Lambda Expressions + Swing + JDBC Basics (accessed April 1, 2026).; Java SE API docs (e.g., `java.util.function`, `javax.swing`, `java.sql`) (accessed April 1, 2026).

Quick Links

Overview Key Topics Demo Summary

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Agenda

1 Overview

2 Key Topics

3 Demo / Example

4 Summary

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Lecture Snapshot

- **Unit focus:** Generics, Lambdas, Swing, and JDBC

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Lecture Snapshot

- **Unit focus:** Generics, Lambdas, Swing, and JDBC
- **Course thread:** Use Swing Basics (Components, Containers, Layouts) to move Campus Resource Manager toward a real desktop application with UI, callbacks, and persistence.

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Recall Before You Start

- **Classes and packages**: Swing applications still depend on clear classes and package boundaries. Main reference: `Unit 02 / Lecture 03 slides/notes`.

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Recall Before You Start

- **Classes and packages**: Swing applications still depend on clear classes and package boundaries. Main reference: `Unit 02 / Lecture 03 slides/notes`.
- **Nested helpers**: GUI code often uses callback and helper patterns introduced earlier. Main reference: `Unit 03 / Lecture 01 slides/notes`.

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Learning Outcomes

- Explain Swing component model: `JFrame`, `JPanel`, components vs containers.

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Learning Outcomes

- Explain Swing component model: `JFrame`, `JPanel`, components vs containers.
- Use layout managers (Flow/Border/Grid) to structure a UI.

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Learning Outcomes

- Explain Swing component model: `JFrame`, `JPanel`, components vs containers.
- Use layout managers (Flow/Border/Grid) to structure a UI.
- Build and run a small multi-panel UI skeleton.

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Key Terms to Know

- **Component**: UI element like button, label, text field.

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- **Component**: UI element like button, label, text field.
- **Container**: holds components (e.g., `JPanel`, `JFrame` content pane).
- **Layout manager**: decides how components are arranged (sizes/positions).
- **Event Dispatch Thread (EDT)**: Swing UI thread (advanced but important for correctness).
- **MVC idea**: separate UI from business logic (we will use this later with JDBC).

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Study Flow for This Lecture

1. Refresh the prerequisite bridge before reading new ideas in isolation.

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1. Refresh the prerequisite bridge before reading new ideas in isolation.
2. Study the main build in this order: *Swing overview: `JFrame`, `JPanel`, components vs containers* → *Layout managers (Flow/Border/Grid)* → *UI composition: multi-panel skeleton*.

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3. Trace the worked example and connect each line back to one concept frame.

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3. Trace the worked example and connect each line back to one concept frame.
4. Attempt the mini exercises before moving to the recap and exit check.

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Unit 04: Generics, Lambdas, GUI Swing & Database Connectivity

- Lecture focus: Swing Basics (Components, Containers, Layouts)

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Today: Roadmap

- Swing overview; ‘JFrame‘, ‘JPanel‘, components vs containers

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Today: Roadmap

- Swing overview; ‘JFrame‘, ‘JPanel‘, components vs containers
- Layout managers (Flow/Border/Grid) and UI composition

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Today: Roadmap

- Swing overview; ‘JFrame‘, ‘JPanel‘, components vs containers
- Layout managers (Flow/Border/Grid) and UI composition
- Building a small multi-panel UI skeleton

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Components vs containers

- Components: `JLabel`, `JButton`, `JTextField`, `JTable`, ...

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Components vs containers

- Components: `JLabel`, `JButton`, `JTextField`, `JTable`, ...
- Containers: `JPanel`, `JFrame` content pane

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Components vs containers

- Components: `JLabel`, `JButton`, `JTextField`, `JTable`, ...
- Containers: `JPanel`, `JFrame` content pane
- Compose UI by nesting panels and adding components inside them.

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Layout managers (quick)

- `FlowLayout`: left-to-right (good for button rows)

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Layout managers (quick)

- `FlowLayout`: left-to-right (good for button rows)
- `BorderLayout`: NORTH/SOUTH/EAST/WEST/CENTER (good for screens)

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Layout managers (quick)

- `FlowLayout`: left-to-right (good for button rows)
- `BorderLayout`: NORTH/SOUTH/EAST/WEST/CENTER (good for screens)
- `GridLayout`: rows/cols (good for simple forms)

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A common UI composition pattern

- Root panel: `BorderLayout`

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A common UI composition pattern

- Root panel: `BorderLayout`
- NORTH: title / header

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A common UI composition pattern

- Root panel: `BorderLayout`
- NORTH: title / header
- CENTER: main panel (form/table)

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A common UI composition pattern

- Root panel: `BorderLayout`
- NORTH: title / header
- CENTER: main panel (form/table)
- SOUTH: action buttons

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Checkpoint and Reflection

Reflection prompt

Where would *Swing Basics (Components, Containers, Layouts)* matter most in Campus Resource Manager, and which design choice would you justify using this lecture's ideas?

Quick self-check

Which part needs one more example before you move on: *Swing overview: `JFrame`, `JPanel`, components vs containers*, *Layout managers (Flow/Border/Grid)*, the worked example, or the recap?

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Example: multi-panel skeleton (1/3)

```
import javax.swing.*;
import java.awt.*;

public class SimpleFormUI {
  public static void main(String[] args) {
    SwingUtilities.invokeLater(() -> {
      JFrame f = new JFrame("CRM Skeleton");
      f.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);

      JPanel root = new JPanel(new BorderLayout(10, 10));
```

Sources: Schildt, *Java: The Complete Reference* (12e) (generics, lambdas, Swing, JDBC).; Bloch, *Effective Java* (3e) (generics + lambdas).; Oracle Java Tutorials: Generics + Lambda Expressions + Swing + JDBC Basics (accessed April 1, 2026).; Java SE API docs (e.g., `java.util.function`, `javax.swing`, `java.sql`) (accessed April 1, 2026).

Example: multi-panel skeleton (2/3)

```
// continued...
    root.add(new JLabel("Add Resource", SwingConstants.
        CENTER),
            BorderLayout.NORTH);

    JPanel form = new JPanel(new GridLayout(2, 2, 8, 8));
    form.add(new JLabel("Name:"));
    form.add(new JTextField());
    form.add(new JLabel("Type:"));
    form.add(new JTextField());
    root.add(form, BorderLayout.CENTER);
```

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Example: multi-panel skeleton (3/3)

```
// continued...
      JPanel actions = new JPanel(new FlowLayout(FlowLayout.
          RIGHT));
      actions.add(new JButton("Save"));
      actions.add(new JButton("Exit"));
      root.add(actions, BorderLayout.SOUTH);

      f.setContentPane(root);
      f.pack();
      f.setLocationRelativeTo(null);
      f.setVisible(true);
    });
  }
}
```

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Mini Exercises (2–3 mins)

1. Which is a container: `JPanel` or `JButton`?

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Mini Exercises (2–3 mins)

1. Which is a container: `JPanel` or `JButton`?
2. Which layout is best for header + body + footer screens?

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Mini Exercises (2–3 mins)

1. Which is a container: `JPanel` or `JButton`?
2. Which layout is best for header + body + footer screens?
3. Why prefer `pack()` over fixed `setSize()` (often)?

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Watch-outs

- Putting every control directly on one JFrame without structure.

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Watch-outs

- Putting every control directly on one JFrame without structure.
- Hardcoding coordinates instead of using layout managers intentionally.

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Watch-outs

- Putting every control directly on one JFrame without structure.
- Hardcoding coordinates instead of using layout managers intentionally.
- Mixing UI code with business logic too early.

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Key Takeaways

- Swing UI = components inside containers.

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Key Takeaways

- Swing UI = components inside containers.
- Layout managers create robust UIs without hard-coded positions.

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Key Takeaways

- Swing UI = components inside containers.
- Layout managers create robust UIs without hard-coded positions.
- Multi-panel composition scales better than one giant panel.

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Checkpoint Questions

1. What is the difference between a component and a container?

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Checkpoint Questions

1. What is the difference between a component and a container?
2. Name 3 layout managers and one use case for each.

Sources: Schildt, *Java: The Complete Reference* (12e) (generics, lambdas, Swing, JDBC).; Bloch, *Effective Java* (3e) (generics + lambdas).; Oracle Java Tutorials: Generics + Lambda Expressions + Swing + JDBC Basics (accessed April 1, 2026).; Java SE API docs (e.g., `java.util.function`, `javax.swing`, `java.sql`) (accessed April 1, 2026).

Checkpoint Questions

1. What is the difference between a component and a container?
2. Name 3 layout managers and one use case for each.
3. Describe one UI screen using the NORTH/CENTER/SOUTH pattern.

Sources: Schildt, *Java: The Complete Reference* (12e) (generics, lambdas, Swing, JDBC).; Bloch, *Effective Java* (3e) (generics + lambdas).; Oracle Java Tutorials: Generics + Lambda Expressions + Swing + JDBC Basics (accessed April 1, 2026).; Java SE API docs (e.g., `java.util.function`, `javax.swing`, `java.sql`) (accessed April 1, 2026).

Next Study Steps

- Revisit the recall references if any older idea still feels weak.

Sources: Schildt, *Java: The Complete Reference* (12e) (generics, lambdas, Swing, JDBC).; Bloch, *Effective Java* (3e) (generics + lambdas).; Oracle Java Tutorials: Generics + Lambda Expressions + Swing + JDBC Basics (accessed April 1, 2026).; Java SE API docs (e.g., `java.util.function`, `javax.swing`, `java.sql`) (accessed April 1, 2026).

Next Study Steps

- Revisit the recall references if any older idea still feels weak.
- Rework the lecture example without looking at the notes first.

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Next Study Steps

- Revisit the recall references if any older idea still feels weak.
- Rework the lecture example without looking at the notes first.
- Attempt the self-study practice and then answer the exit questions in full sentences.

Sources: Schildt, *Java: The Complete Reference* (12e) (generics, lambdas, Swing, JDBC).; Bloch, *Effective Java* (3e) (generics + lambdas).; Oracle Java Tutorials: Generics + Lambda Expressions + Swing + JDBC Basics (accessed April 1, 2026).; Java SE API docs (e.g., `java.util.function`, `javax.swing`, `java.sql`) (accessed April 1, 2026).

Next Study Steps

- Revisit the recall references if any older idea still feels weak.
- Rework the lecture example without looking at the notes first.
- Attempt the self-study practice and then answer the exit questions in full sentences.
- Apply the idea once inside Campus Resource Manager to make the concept stick.

Sources: Schildt, *Java: The Complete Reference* (12e) (generics, lambdas, Swing, JDBC).; Bloch, *Effective Java* (3e) (generics + lambdas).; Oracle Java Tutorials: Generics + Lambda Expressions + Swing + JDBC Basics (accessed April 1, 2026).; Java SE API docs (e.g., `java.util.function`, `javax.swing`, `java.sql`) (accessed April 1, 2026).

Next Lecture Preview

Next: Swing event handling + JDBC (persistence)

- We will connect buttons to actions and store data in a database.

Sources: Schildt, *Java: The Complete Reference* (12e) (generics, lambdas, Swing, JDBC).; Bloch, *Effective Java* (3e) (generics + lambdas).; Oracle Java Tutorials: Generics + Lambda Expressions + Swing + JDBC Basics (accessed April 1, 2026).; Java SE API docs (e.g., `java.util.function`, `javax.swing`, `java.sql`) (accessed April 1, 2026).